

OilRig (APT34)

Cyber Threat Actor Analysis

Cybersecurity Final Project

Threat Actor Classification

- Advanced Persistent Threat (APT)
- Aliases: APT34, Helix Kitten, Crambus, Cobalt Gypsy
- Long-term cyber espionage campaigns

Origin and Capabilities

- Believed linked to Iranian intelligence services
- Active since around 2014
- Highly skilled operators with government resources

Primary Targets

- Energy and oil companies
- Financial institutions
- Telecommunications providers
- Government organizations

Motivations

- Strategic cyber espionage
- Gather intelligence on political rivals
- Support Iranian geopolitical objectives

Reconnaissance & Weaponization

- Research target employees and infrastructure
- Develop custom malware tools
- Prepare spear■phishing campaigns

Delivery & Exploitation

- Spearphishing emails
- Credential harvesting attacks
- Exploitation of system vulnerabilities

Installation & Command Control

- Deploy backdoors on compromised systems
- Maintain persistence inside networks
- Communicate with command and control servers

Case Study Impact

- Energy sector cyber espionage operations
- Primary effects: compromised systems
- Secondary effects: financial loss and disruption
- Second-order effects: regional security concerns

Policy Implications

- Represents a public cybersecurity concern
- Requires international cyber defense cooperation
- Organizations must invest in threat intelligence and security awareness